

Discussion of

Time-Series Momentum across the Capital Structure

Gruenthaler, Manconi, de Roon and Tancheva

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The Big Picture — Motivation

$$\text{Momentum}_t = \mathcal{F}_t[\text{Risk}]$$

- Momentum is the anomaly that has most resisted a risk-based account
- Fama–French (1996) conceded their three-factor model cannot explain it
- Rational accounts need a mechanism that makes expected returns MOVE with past returns
 - Johnson (2002), Sagi–Seasholes (2007), Garlappi–Yan (2011), Liu–Zhang (2008)
- The difficulty: in most models a price fall makes a firm RISKIER, which gives reversal

So the question is not whether momentum is risky. It is what makes a fallen firm's expected return go DOWN.

The Background for this Paper

- Equity is a call option on firm assets, struck at the face value of debt (Merton, 1974)

$$E = A \Phi(d_1) - Fe^{-r_f \tau} \Phi(d_2)$$

- Expected return follows from the elasticity of equity to asset value:

$$g = r_f + \eta^A (\mu - r_f), \quad \eta^A = \Phi(d_1) \frac{A}{E} \geq 1$$

- A price fall raises leverage $\rightarrow$ raises $\eta^A \rightarrow$ raises expected returns $\rightarrow$ REVERSAL
 - Their own Table A-equivalent: pure Merton simulation gives a winner-minus-loser spread with the wrong sign

Fifty years of structural credit modelling, and nobody used Merton to explain momentum — because Merton predicts the opposite.

This Paper in a Nutshell

$$g = r_f + \underbrace{\eta^A(\mu - r_f)}_{\text{leverage} \rightarrow \text{reversal}} + \underbrace{\eta^V \lambda_V}_{\text{variance} \rightarrow \text{continuation}}$$

$$\eta^A = E_A \frac{A}{E} > 0, \quad \eta^V = E_V \frac{V}{E} > 0, \quad \lambda_V < 0$$

- Add a SECOND priced state: the variance input V, with its own premium λ
 - Equity is a call, so it is long vega: $\eta^V > 0$
 - Variance risk is negatively priced: $\lambda < 0$ (Carr–Wu 2009; Bollerslev–Tauchen–Zhou 2009)
 - A fallen stock is more option-like $\rightarrow$ more vega $\rightarrow$ LOWER expected return $\rightarrow$ continuation

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- Momentum arises where the second dominates — as a condition on a covariance:

$$\text{Cov}_t(dg, r) > 0 \iff \text{momentum}$$

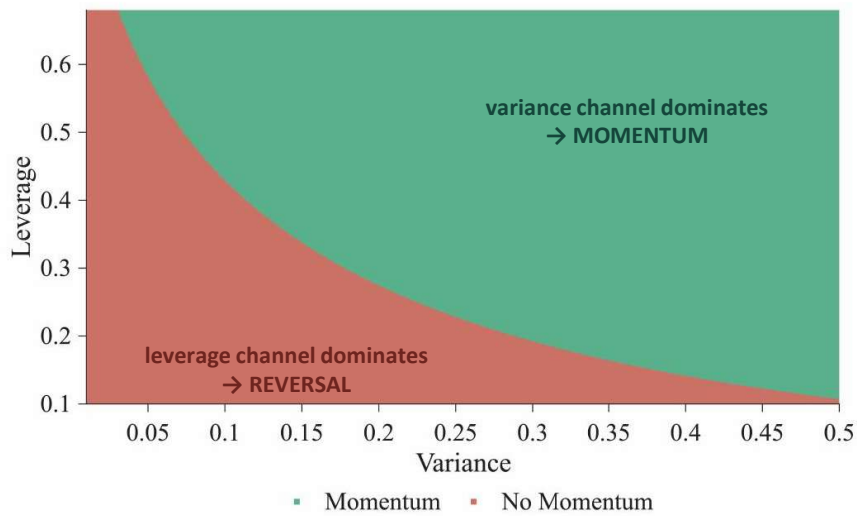
$$\frac{\text{Cov}_t(dg, r)}{dt} = \frac{1}{E} \left[g_A E_A \sigma_A^2 A^2 + g_V E_V \sigma_V^2 V + \rho \sigma_A \sigma_V A \sqrt{V} (g_A E_V + g_V E_A) \right]$$

- $\text{Cov} > 0 \rightarrow \text{momentum}$; $\text{Cov} < 0 \rightarrow \text{REVERSAL}$. Both are directional predictions, and both are tested
 - $\approx 12\%$ of firm-months satisfy it

The contribution is not that momentum is stronger somewhere — it is a SIGN FLIP at a specific point in (leverage, variance) space.

This Paper in a Nutshell

- Your Figure 1: the sign of $\text{Cov}(dg, r)$ over leverage and variance, at the calibrated parameters



- **Green: $\text{Cov} > 0$**
 - $\eta^v \lambda$ outweighs $\eta^A(\mu - r_f)$. A price fall raises vega faster than it raises asset-risk exposure
- **Red: $\text{Cov} < 0$**
 - The Merton force wins — a price fall raises leverage and expected returns
- **The boundary is a TRADE-OFF**
 - More variance buys momentum at lower leverage: the frontier falls from ≈ 0.55 at $V = 0.05$ to ≈ 0.11 at $V = 0.50$

- **The same condition, applied across the capital structure:**
 - Debt is concave $\Rightarrow \zeta^v < 0 \Rightarrow$ bonds REVERSE; shared variance exposure $\Rightarrow$ equity-to-bond SPILLOVER
 - Aggregated into winner and loser portfolios $\Rightarrow \Delta^v < 0 \Rightarrow$ momentum CRASHES

Outline

The Paper

My Comments

Final Remarks

4 comments, plus smaller points and possible checks

1) What Is the Second State Variable Doing?

- Plain Merton already has variance — but as a **PARAMETER**, not a state

- σ_A is fixed. Equity volatility $\eta^A \sigma_A$ does rise when the price falls, but that is leverage, not a second risk
- One priced risk, so expected return is monotone in leverage $\Rightarrow$ REVERSAL, always

$$g^{\text{Merton}} = r_f + \eta^A (\mu - r_f), \quad \sigma_A \text{ constant}$$

- Your addition is not that **V moves**. It is that **V is PRICED**

- V must be non-spanned — variance shocks cannot be replicated by trading the asset — so it earns its own premium λ
- That premium is negative, which is what flips the sign

Setting $\lambda = 0$ keeps V stochastic but UNPRICED	$\lambda = -1.75$	$\lambda = -1.00$	$\lambda = -0.50$	$\lambda = 0$
Simulated firms with Cov > 0 (your Table 1 design)	32.45%	21.41%	3.78%	0.00%

So it is not the stochasticity that matters — it is the pricing. A second state that moves but is unpriced adds essentially nothing.

1) The Negative Price of Variance Risk

- The paper takes $\lambda < 0$ as given — cited to Carr–Wu (2009) and Bollerslev–Tauchen–Zhou (2009), calibrated from Du–Elkamhi–Ericsson
 - That is settled at the INDEX level — but contested at the firm level, where λ operates:
 - **Negatively priced:** Zhang–Zhou–Zhu (2009, RFS) · Cao–Han (2013, JFE) · Du–Elkamhi–Ericsson (2019, JF) · Duarte–Jones–Wang (2024, JF)
 - **Unpriced or positive:** Merton (1987, JF) · Goyal–Santa-Clara (2003, JF) · Driessen–Maenhout–Vilkov (2009, JF) · Fu (2009, JFE) · Stambaugh–Yu–Yuan (2015, JF)

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The literature	What it establishes
Barinov (2018, JFE) § 4.2	equity as a call on assets; distress via O-score; FVIX absorbs a 1.027 ($t = 3.28$) alpha to -0.002
Barinov & Chabakauri (2023, RAPS)	the model — option-like, high-IVOL firms as hedges against aggregate volatility risk

- Barinov:** distressed, high-volatility firms $\Rightarrow$ “a call option on the assets” $\Rightarrow$ high FVIX betas $\Rightarrow$ **HEDGES** $\Rightarrow$ low expected returns
 - These firms have a large positive loading on a factor whose price we know is negative.
- Your paper:** These are the same firms that need the negative variance price of risk to generate momentum.
- The connection:** yours is the **DYNAMIC** version of his cross-section — price falls $\Rightarrow$ more option-like (FVIX beta rises) $\Rightarrow$ expected return falls $\Rightarrow$ **CONTINUATION**

A microfoundation derived for levered, distressed, option-like equity — engaging with it turns an assumed, and contested, parameter into a mechanism.

1) A Mapping Worth Making

- If both frameworks describe the same compensation, the two should line up firm by firm:

$$\underset{\text{model-implied}}{\eta_i^V \cdot \lambda_V} \longleftrightarrow \underset{\text{estimated from returns}}{\beta_{FVIX, i} \cdot \lambda_{FVIX}}$$

- **The direct check: does η^V predict β_{FVIX} in the cross-section?**
 - If it does, your structural quantity is capturing what the factor literature measures — an independent validation of the calibration, and of the mechanism
 - Two further things the same construction settles: does the Cov > 0 alpha survive adding FVIX to Table 4, and does Δ^V predict crashes beyond an FVIX loading?
- **And the mapping has a MODEL version — pricing only the systematic part of V**
 - $dV_i = \beta_i \cdot dV_M + \text{idio}$, with only V_M priced $\Rightarrow \lambda_i = \beta_i \cdot \lambda_M$ — the firm-level sign then follows from the index premium, which is not contested
 - Closed forms survive — λ only rescales g_V ; and β_i is estimable from the CIV factor (Herskovic et al.) for the FULL sample, not just post-1990
 - High-vega, LOW- β firms become predicted no-momentum firms — a new falsifiable margin the assumed- λ version does not have

Either outcome helps — and you could have an independent validation of the calibration and the mechanism.

2) Something Else the Model Implies

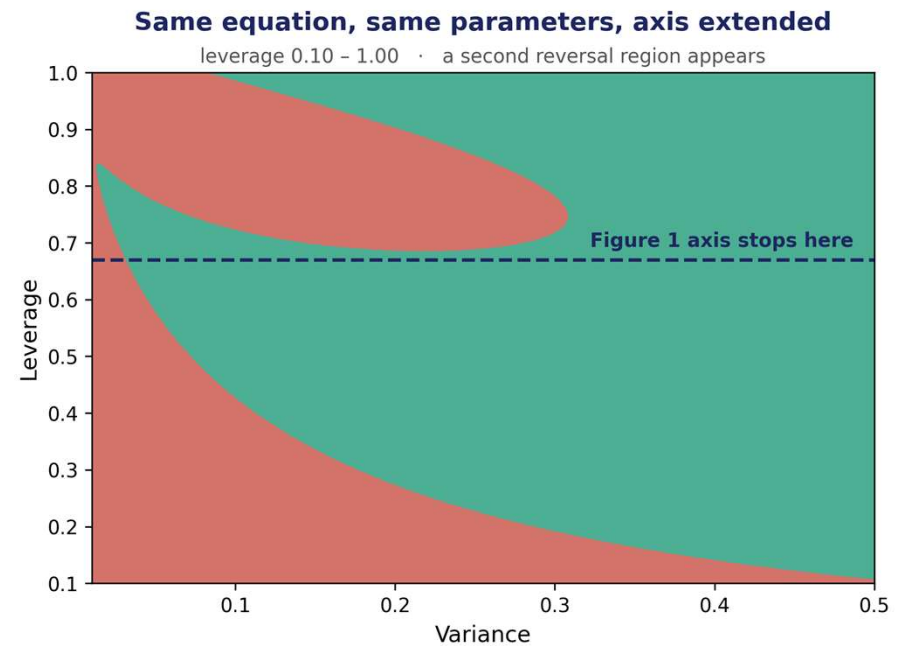
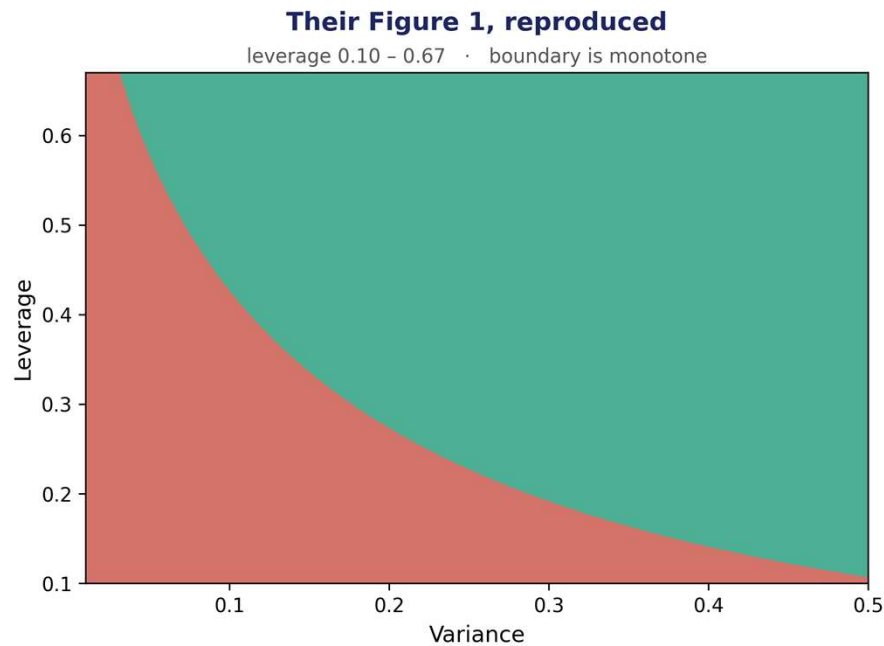
- Evaluating equation (11) at your parameters over a wider leverage range than Figure 1 plots gives something interesting:

Input volatility	Sign changes in leverage	Implied structure
$\sqrt{V} = 0.20$	0.62 and 0.79	reversal → momentum → reversal
$\sqrt{V} = 0.30$	0.45, 0.73, 0.99	reversal → momentum → reversal → momentum
$\sqrt{V} = 0.39$	0.34, 0.70, 0.95	reversal → momentum → reversal → momentum

- Figure 1's leverage axis stops at 0.67 — just below the second crossing. Figure 2 already plots leverage to 1.0
 - Verified: crossings identical at step sizes $1e-3$ to $1e-6$ and across two implementations; vega matches your closed form to $1e-10$; signs correct in every region you describe
- The economics of the upper region:
 - Through ASSET shocks the variance channel gives continuation — a fall raises leverage, raises vega, lowers expected returns
 - Through VARIANCE shocks at high leverage it gives REVERSAL — volatility spikes, the near-the-money equity jumps, and its expected return collapses at the same moment

The paper says the variance channel generates continuation. Your equation says its sign depends on how option-like the equity has become — a richer prediction, and one no reduced-form volatility story can make.

2) The Same Equation, Plotted Further



- Liquidity or behavioural explanation would predict monotone ordering. You: unique hump pattern.
- In my classification $P(\text{Cov} > 0)$ rises to 54% at leverage 0.65–0.75, then falls to 34% above 0.75
- This could also relate to highly leverage distressed losers generating **momentum crashes**.

2) Worth Exploring

- **Test the upper boundary as you already test the lower one**
 - Table 6 tests the low-leverage end and confirms reversal. The symmetric test at high leverage is the same exercise
 - If reversal appears, it is a prediction no factor loading could produce — it strengthens you against comment 1 too
- **Reporting the sensitivity of the PARTITION, not just the spread**
 - Table IA.1 reports only the $h = 6$ Diff — never how the fraction satisfying or the boundary responds
 - Worth reconciling: IA.1 shows the Diff moving 9% as λ goes $-0.5 \rightarrow -3$, but I compute $P(\text{Cov} > 0)$ at 3.78% versus 32.45% over that range
- **The second boundary is not a corner — it connects to your Section 5**
 - 27.6% of your own Table 1 design sits in the upper region — so the $\text{Cov} < 0$ column pools TWO reversals; splitting it at the boundary is a free test
 - The region fills in bear markets with the LOSER leg — reversal there is losers rallying on volatility: a momentum crash. Do rebound-month loser rallies concentrate above the boundary?
 - A hump in continuation over distress is Garlappi–Yan’s (2011) signature — reached through recovery, not priced variance
- And one free line in the text: say the variance channel's sign depends on moneyiness

3) An Additional Test Worth Running

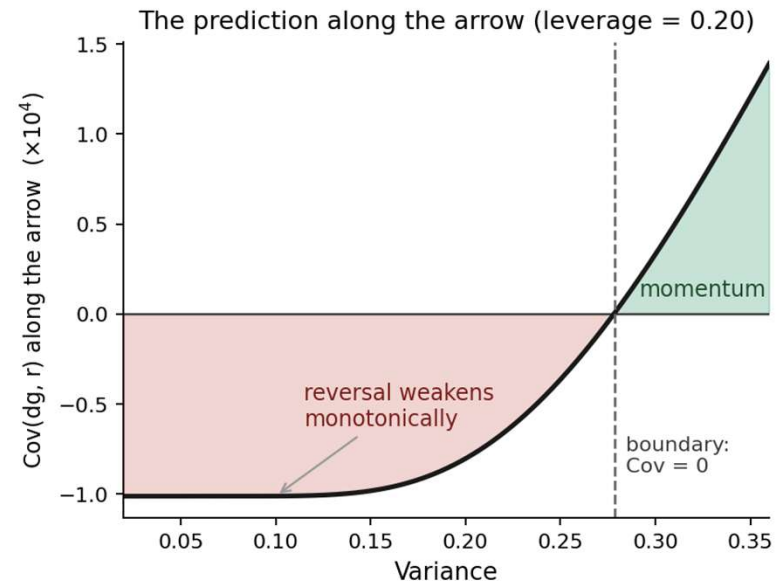
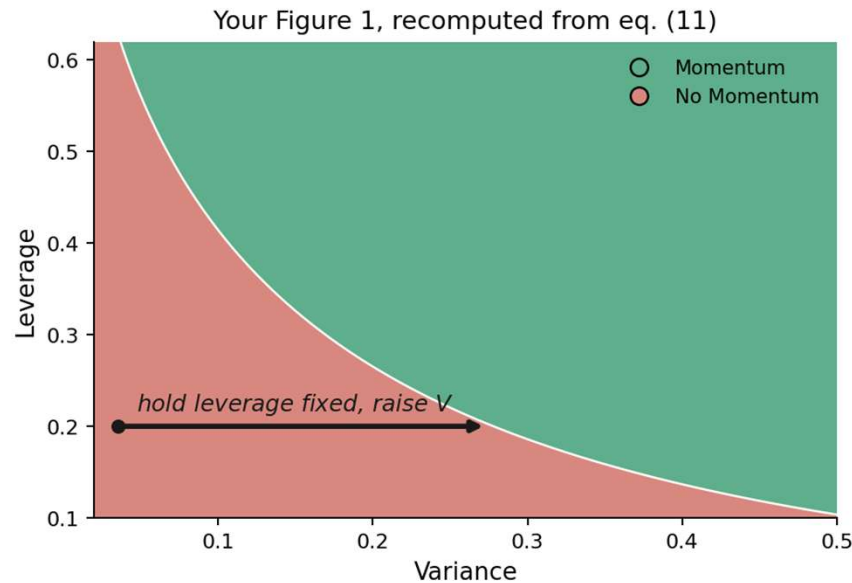
- **Inside your reversal region, the model implies an ORDERING in variance:**
 - Holding leverage fixed, raising V moves a firm TOWARD the boundary — so $|\text{reversal}|$ should WEAKEN as variance rises
 - Table 6 selects on low leverage AND low variance together, so it cannot show this

	Prediction inside the reversal region
Your model (Figure 1 geometry)	reversal STRONGEST at low variance, weakening toward the boundary
Jegadeesh (1990); Nagel (2012); Avramov–Chordia–Goyal (2006)	reversal STRONGEST at high variance — liquidity provision scales with risk

- **Opposite signs on the same dimension — which makes it discriminating**
- **Jegadeesh, Goyal, Nagel: reversal is compensation for absorbing order flow – scaling with risk.**
 - A simple sort on $\sqrt{V}$ within the $\text{Cov} < 0$ group separates your mechanism from the known short-horizon effect
 - If reversal weakens as variance rises, that is evidence FOR you that no reduced-form story predicts — and it is not in the paper

The reversal side currently rests on a subsample chosen to be far from the boundary. The ordering AS you approach it would be a sharper test.

3) The Ordering, Drawn in Figure 1



- The model is quantitative here, not just directional: Cov is flat deep in the region, fades MONOTONICALLY as V approaches the boundary, and never changes sign before it
 - Computed from equation (11) at your calibration — the frontier matches Figure 1: 0.56 at $V = 0.05$, 0.10 at $V = 0.50$

Smaller Points and Possible Robustness Checks

- **The simulation and the data disagree about horizons**
 - Table 1: spread 0.081 at $h = 1$ falling to 0.007 at $h = 12$. Table 3: Diff 0.302 at $h = 1$ rising to 0.841 at $h = 6$. The simulation uses the true innovation, the data a proxy — the gap measures how noisy past returns are as that proxy
 - **The reversal side rests on the adjacent month**
 - Skipping one month takes Table 6 from -0.361 ($t = -3.61$) to $+0.042$ ($t = 0.39$). Not bid–ask bounce — I tested it: a one-day skip leaves it at -0.320 , and it is significant among liquid stocks
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- **Possible robustness checks**
 - Add distance-to-default to Table 8 — same three inputs, different transform, nearly orthogonal to the condition in my data (Spearman 0.15). Your rating control exists only for rated firms, which is thinnest exactly where the condition binds
 - Report the market-cap share and firm counts of the condition groups — the tables are value-weighted, so cap share is what determines the portfolios. A size split would also separate this from Avramov et al. (2007)
 - Value-weight the bond tables (the equity tables are VW); and $\tau = 1$ is the maximum in Table IA.1 — the Diff falls 68% by $\tau = 4$

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- The core contribution is novel and worth developing — a covariance condition producing a sign prediction, applied jointly across the capital structure. Nobody else has that object
- My comments are mostly about connections worth making — to a literature that explains your key assumption, and to implications the model already contains
- **Build FVIX and use it three times — Tables 4, 8 and 15**
- **Extending Figure 1 to leverage 1.0, and testing the upper boundary**
- **Decomposing the Table 10 spillover signal into its asset and variance components — eq. (8) defines the split**

A model that predicts a sign flip at a specific point in characteristic space is doing something no characteristic and no factor loading can do. That is the claim worth making — and worth testing directly.